6 Application in Object Detection

After having analyzed the quasi-stationary proximal ADMM, we want to look at a possible workflow for designing and configurating the algorithm for a specific model, especially solving the subproblems, and practically evaluate the performance of the algorithm. For this purpose, we consider a particular tensor decomposition model in the context of an important field in unsupervised learning, namely object detection in video data. The model proposed by Yin et al. in [20, Sect. II.B] is designed for object detection in satellite image data.

6.1 Introduction to 3D-Tensors

To understand the setting, we briefly explain some fundamentals of 3D-tensors. This section is based on [7], [8] and slightly adapts the notation.

Let $R \in \mathbb{R}^{T \times M \times N}$ be arbitrary. Then $R^{(t)} := R_{t,:,:} \in \mathbb{R}^{M \times N}$ abbreviates the t-th time slice of R and $R_{:,m,n} \in \mathbb{R}^T$ represented as a column vector is referred to as the m-n-fiber of R.

We denote by $r \in \mathbb{R}^{T \cdot M \cdot N}$ the vectorization of R beginning with the time axis, i.e. vectorizing each time slice columnwise by stacking these into a column vector. We define the tensorial H-induced norm for some positive definite matrix $H \in \mathbb{R}^{(T \cdot M \cdot N) \times (T \cdot M \cdot N)}$ in terms of the corresponding H-induced norm for vectors: $\|R\|_H := \|r\|_H = \sqrt{r^T H r}$. The tensorial Frobenius norm is the special case for $H = \mathbb{I}_{T \cdot M \cdot N}$ and corresponds to the Euclidean norm of the vectorization r, and can be written in terms of the matrix Frobenius norm of the time slices of R:

$$\|R\|_F = \|r\|_{\mathbb{I}_{T \cdot M \cdot N}} = \sqrt{\sum_{i=1}^{T \cdot M \cdot N} r_i^2} = \sqrt{\sum_{t=1}^{T} \|R^{(t)}\|_F^2}$$

Let $\overline{R} \in \mathbb{R}^{TxMxN}$ denote the Discrete Fourier Transform (DFT) of R along the time-dimension, i.e. taking each m-n-fiber and applying the FFT individually, which mathematically corresponds to multiplying with the complex Fourier matrix $\mathbb{F}_T$: $\overline{R}_{:,m,n} = \mathbb{F}_T \cdot R_{:,m,n}$. By this representation of the DFT as a fiberwise matrix multiplication, we see that the map $R \mapsto \overline{R}$ is linear.

It holds $\mathbb{F}_T (\mathbb{F}_T)^H = (\mathbb{F}_T)^H \mathbb{F}_T = T \cdot \mathbb{I}_T$ (cf. [9, Eq. (7)]) and therefore $\frac{1}{\sqrt{T}} \mathbb{F}_T$ is unitary. As also the tensorial Frobenius norm is element- and hence fiber-separable, and the Euclidean ℓ_2-norm unitarily invariant, it holds:

$$\|R\|_F^2 = \sum_{m=1}^{M} \sum_{n=1}^{N} \|R_{:,m,n}\|^2 = \sum_{m=1}^{M} \sum_{n=1}^{N} \|\frac{1}{\sqrt{T}} \mathbb{F}_T \cdot R_{:,m,n}\|^2$$

$$= \frac{1}{T} \sum_{m=1}^{M} \sum_{n=1}^{N} \|\mathbb{F}_T \cdot R_{:,m,n}\|^2 = \frac{1}{T} \sum_{m=1}^{M} \sum_{n=1}^{N} \|\overline{R}_{:,m,n}\|^2 = \frac{1}{T} \|\overline{R}\|_F^2$$

Moreover, define $R_{\text{ifft}} \in \mathbb{R}^{T \times M \times N}$ as the inverse DFT of R, i.e. fiberwise $(R_{\text{ifft}})_{:,m,n} := \frac{1}{T} (\mathbb{F}_T)^H \cdot R_{:,m,n}$. Because of $(\mathbb{F}_T)^H \mathbb{F}_T = T \cdot \mathbb{I}_T$, by comparing the m-n-fibers for $m \in [M]_{\geq 1}, n \in [N]_{\geq 1}$, we see that the DFT and inverse DFT are really inverse operations:

$$(\overline{R_{\text{ifft}}})_{:,m,n} = \mathbb{F}_T \cdot (R_{\text{ifft}})_{:,m,n} = \frac{1}{T} \mathbb{F}_T (\mathbb{F}_T)^H \cdot R_{:,m,n} = \mathbb{I}_T \cdot R_{:,m,n} = R_{:,m,n}$$

6.2 Model

We introduce the model from [20, Sect. II]. For the sake of simplicity in the theoretical analysis and algorithm development for the X-subproblem later, we will use the regular tensorial Schatten q-norm scaled by a constant instead of a weighted variant which is used in [20, Eq. (5)].

A video is a sequence of T frames, and each $M \times N$-frame consists of $M \cdot N$ pixels. We assume that all frames of the video are grayscale, so each pixel is encoded by the grayscale-value in $[255]$.
In contrast to simplified models, which transform the frames into column vectors and stack them side by side, the video will be represented as a tensor to preserve the patch-spatial information, especially continuity between frames. (cf. [20, Sect. I]). We denote this video-tensor as $M \in \mathbb{R}^{T \times M \times N}$, since T is the number of frames of the video, and each frame is a $M \times N$-image. Each frame in the video corresponds to a time slice $M_{t,:,:}$ for $t \in [T]_{\geq 1}$.

Yin et al. assume there is some random noise, which also takes in account for turbulence in camera movement and other disturbances and hence might lead to better detection performance as they observe in their numerical experiments. In total the representation of the video can be additively decomposed into the following tensors for the background, the foreground and the noise:

- **Background X**: As the background of each frame is static and in many cases homogeneous, a low-rank model for the signal components at a certain frequency in the Fourier domain is assumed, i.e. each slice $\overline{X}^{(t)}$ of the DFT of X is low-rank.

  A good proxy for the matrix-rank has been found in the Schatten q-norm (cf. [8]):

  $$A \mapsto \|A\|_{C_X,q} = \left(\sum_{i=1}^{min(M,N)} C_X \cdot \sigma_i(A)^q \right)^{\frac{1}{q}}$$

  especially for smaller values $q \in (0,1)$, which in turn make the Schatten q-norm nonconvex. A generalization of the Schatten q-norm to tensors in the Fourier domain is given by [8, Def. 6]:

  $$A \mapsto \|A\|_{C_X,q} = \left(\frac{1}{T} \sum_{t=1}^{T} \|\overline{A}^{(t)}\|_{C_X,q}^q \right)^{\frac{1}{q}}$$

- **Foreground Y**: Since the object of interest is typically small in comparison to the magnitude of the video environment, the foreground, i.e. the object, is assumed to only take up a small amount of space in each frame and accordingly the corresponding time slices $Y^{(t)}$ of Y to be sparse. A good model for this heuristic is given by the l_1-norm.

- **Noise Z**: The random noise Z will be modeled by the squared tensorial Frobenius norm.

The derived decomposition priors can be formulated as an optimization problem, which is the model in [20, Eq. (7)] for some $q \in (0,1]$:

$$\begin{aligned} \min_{X,Y,Z} \quad & \|X\|_{C_X,q}^q + \lambda \|Y\|_1 + \mu \|Z\|_F^2 \\ \text{subject to} \quad & X + Y + Z = M \end{aligned}$$

By taking the tensorial inner product $\langle K^1, K^2 \rangle = \sum_{t=1}^{T} \sum_{m=1}^{M} \sum_{n=1}^{N} K^1_{t,m,n} \cdot K^2_{t,m,n}$, which induces the tensorial Frobenius norm, the corresponding augmented Lagrangian has the following form:

$$\mathcal{L}_\beta(X,Y,Z,W) = \|X\|_{C_X,q}^q + \lambda \|Y\|_1 + \mu \|Z\|_F^2 + \langle W, X + Y + Z - M \rangle + \frac{\beta}{2} \|X + Y + Z - M\|^2$$

To reduce the subproblems to proximal mappings, we will choose $H_X = H_Y = \beta \cdot \mathbb{I}_{T \cdot M \cdot N} \succ 0$ for the proximal terms in the X- and Y-subproblem, so $\|\cdot\|_{H_X}^2 = \|\cdot\|_{H_Y}^2 = \beta\|\cdot\|_F^2$, and regard the completed form.

Notice the following for any tensors u, s_1, $s_2 \in \mathbb{R}^{T \times M \times N}$ by expanding the squared tensorial Frobenius norm, followed by collecting equal terms and completing the square:

$$\begin{aligned}
\|u - s_1\|_F^2 + \|u - s_2\|_F^2 &= \|u\|_F^2 - 2\langle u, s_1\rangle + \|s_1\|_F^2 + \|u\|_F^2 - 2\langle u, s_2\rangle + \|s_2\|_F^2 \\
&= 2\|u\|_F^2 - 2\langle u, s_1 + s_2\rangle + \|s_1\|_F^2 + \|s_2\|_F^2 \\
&= 2\|u\|_F^2 - 2 \cdot 2\langle u, \frac{1}{2}(s_1 + s_2)\rangle + \|\frac{1}{2}(s_1 + s_2)\|_F^2 + \|s_1\|_F^2 + \|s_2\|_F^2 - \|\frac{1}{2}(s_1 + s_2)\|_F^2 \\
&= 2\|u - \frac{1}{2}(s_1 + s_2)\|_F^2 + \|s_1\|_F^2 + \|s_2\|_F^2 - \|\frac{1}{2}(s_1 + s_2)\|_F^2
\end{aligned}$$

Hence, by taking $u = X$, $s_1 = M - Y^k - Z^k - W^k/\beta$ and $s_2 = X_k$, and then discarding terms with the constants s_1 and s_2 with respect to X, the X-subproblem in the completed form reduces to:

$$\begin{aligned}
X^{k+1} &\approx \arg\min_X \ \|X\|_{C_X,q}^q + \frac{\beta}{2}\|X + Y + Z - M + W^k/\beta\|_F^2 + \frac{\beta}{2}\|X - X^k\|_F^2 \\
&= \arg\min_X \ \|X\|_{C_X,q}^q + \frac{\beta}{2}\|X - \big(M - Y - Z - W^k/\beta\big)\|_F^2 + \frac{\beta}{2}\|X - X^k\|_F^2 \\
&= \arg\min_X \ \|X\|_{C_X,q}^q + \beta \cdot \|X - \frac{1}{2}\big(M - Y^k - Z^k - W^k/\beta + X^k\big)\|_F^2
\end{aligned}$$

Similarly, the Y-Subproblem reduces to:

$$\begin{aligned}
Y^{k+1} &\approx \arg\min_Y \ \lambda\|Y\|_1 + \frac{\beta}{2}\|X + Y + Z - M + W^k/\beta\|_F^2 + \frac{\beta}{2}\|Y - Y^k\|_F^2 \\
&= \arg\min_Y \ \lambda\|Y\|_1 + \beta \cdot \|Y - \frac{1}{2}\big(M - X^{k+1} - Z^k - W^k/\beta + Y^k\big)\|_F^2
\end{aligned}$$

Algorithm 4 Generic Quasi-Stationary Proximal ADMM for Object Detection

Initialize X^0, Y^0, Z^0, W^0
while stopping criteria not fulfilled **do**
  $X^{k+1} \approx \arg\min_X \ \|X\|_{C_X,q}^q + \beta \cdot \|X - \frac{1}{2}(M - Y^k - Z^k - W^k/\beta + X^k)\|_F^2$
  $Y^{k+1} \approx \arg\min_Y \ \lambda\|Y\|_1 + \beta \cdot \|Y - \frac{1}{2}(M - X^{k+1} - Z^k - W^k/\beta + Y^k)\|_F^2$
  $Z^{k+1} \approx \arg\min_Z \ \mu\|Z\|_F^2 + \frac{\beta}{2} \cdot \|Z - (M - X^{k+1} - Y^{k+1} - W^k/\beta)\|_F^2$
  $W^{k+1} \longleftarrow W^k + \beta(X^{k+1} + Y^{k+1} + Z^{k+1} - M)$
  $k \longleftarrow k + 1$
end while
return X^k, Y^k, Z^k, W^k

Therefore, the subproblems are reduced to just proximal mappings. Our next goal is to formally derive and develop the sketched subschemes from [20] to efficiently solve these subproblems. We examine more generally the proximal operators given by each subproblem.

The X-subproblem demands a more technical treatment, but through some considerations can be reduced to solving many instances of a nonconvex nonsmooth problem in one dimension, which is parallelizable. We will see that the strongly convex Y- and Z-subproblems possess closed-form solutions. The proximal operator of the l_1-norm is well-known and reduces to soft thresholding, while the Z-subproblem can be solved by using the necessary and sufficient optimality condition for convex minimization, that is stationarity with respect to the convex subdifferential.

6.3 Configuration of Subproblems

The X-Subproblem

Let us determine the proximal operator of the tensorial Schatten q-norm for $\alpha > 0$. Fix an arbitrary tensor A. The proximal function is given by:

$$\phi_X(R) = \|R\|_{C_X,q}^q + \frac{\alpha}{2}\|R - A\|_F^2$$

Let R be some tensor. Let us collect minimizers $X^{(t)}$ of the proximal matrix mappings

$$H^t \mapsto \|H^t\|_{C_X,q}^q + \frac{\alpha}{2}\|H^t - \overline{A}^{(t)}\|_F^2$$

for $t \in [T]_{\geq 1}$ into a tensor X along the time-dimension with $\overline{A}^{(t)}$ being the t-th time slice of the DFT of A. Therefore:

$$\forall t \in [T]_{\geq 1} : \|\overline{R}^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|\overline{R}^{(t)} - \overline{A}^{(t)}\|_F^2 \geq \|X^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|X^{(t)} - \overline{A}^{(t)}\|_F^2$$

We can rewrite $\phi_X(R)$ using the definition of the tensorial Schatten q-norm, the norm-correspondence for the DFT, its linearity and the slicewise separability of the Frobenius norm:

$$\begin{aligned}\phi_X(R) &= \|R\|_{C_X,q}^q + \frac{\alpha}{2}\|R - A\|_F^2 = \frac{1}{T}\sum_{t=1}^{T}\|\overline{R}^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2T}\|\overline{R - A}\|_F^2 \\ &= \frac{1}{T}\left(\sum_{t=1}^{T}\|\overline{R}^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|\overline{R} - \overline{A}\|_F^2\right) = \frac{1}{T}\sum_{t=1}^{T}\left(\|\overline{R}^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|\overline{R}^{(t)} - \overline{A}^{(t)}\|_F^2\right)\end{aligned}$$

By construction, each summand is minimized by X, so it holds:

$$\phi_X(R) = \frac{1}{T}\sum_{t=1}^{T}\left(\|\overline{R}^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|\overline{R}^{(t)} - \overline{A}^{(t)}\|_F^2\right) \geq \frac{1}{T}\sum_{t=1}^{T}\left(\|X^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|X^{(t)} - \overline{A}^{(t)}\|_F^2\right)$$

By bijectivity of the DFT we have $\overline{X_{\text{ifft}}}^{(t)} = X^{(t)}$, and therefore we conclude that X_{ifft} is a solution to the proximal problem:

$$\phi_X(R) = \frac{1}{T}\sum_{t=1}^{T}\left(\|\overline{R}^{(t)}\|_{C_X,q}^q + \frac{\alpha}{2}\|\overline{R}^{(t)} - \overline{A}^{(t)}\|_F^2\right) \geq \frac{1}{T}\sum_{i=1}^{T}\left(\|\overline{X_{\text{ifft}}}^{(i)}\|_{C_X,q}^q + \frac{\alpha}{2}\|\overline{X_{\text{ifft}}}^{(i)} - \overline{A}^{(i)}\|_F^2\right) = \phi_X(X_{\text{ifft}})$$

By our considerations, the tensorial Schatten q-norm proximal problem is solved by building the DFT of A, computing the matrix proximal operator of the Schatten q-norm slicewise, concatenating the solutions along the time dimension into a tensor X, and at last applying the inverse DFT. The DFT and inverse DFT can be computed efficiently by the Fast-Fourier-Transform (FFT).

To efficiently compute the second substep, we need to examine the matrix proximal operator of the Schatten q-norm. We generalize a result from Bauschke and Combettes for proximal operators of convex matrix functions that depend only on the singular values (cf. [3, Prop. 24.68]). The result is based on the von Neumann trace inequality (a concise proof can be found in [10]), and extend it to complex proximal minimization of possibly nonconvex functions:

Proposition 5. *Let $\alpha > 0$. Set $m = \min\{M, N\}$ and $I = [m]_{\geq 1}$, and suppose ϕ to be a proper, even function on $\mathbb{R}$. Set*

$$f : \mathbb{C}^{MxN} \to (-\infty, +\infty] : A \mapsto \sum_{i=1}^{m}\phi(\sigma_i(A))$$

Now let $A \in \mathbb{C}^{MxN}$, let $U \in \mathbb{U}^M$, $\Sigma \in \mathbb{R}^{MxN}$ and $V \in \mathbb{U}^N$ be such that $A = U\Sigma V^T$ is a singular value decomposition of A. Define

$$\Sigma_f = \mathrm{Diag}\left(|Prox_\phi^\alpha(\sigma_1(A))|, \ldots, |Prox_\phi^\alpha(\sigma_m(A))|\right)$$

Then

$$Prox_f^\alpha(A) = U\Sigma_f V^T$$

Proof. See Appendix. □

Choosing $\phi(t) = C_X|t|^q$ and $\alpha = 2\beta$ recovers $\|\cdot\|_{C_X,q}^q$ as the function f and satisfies the conditions of Proposition 5:

$$\phi(t) = C_X|t|^q = C_X|-t|^q = \phi(-t) \quad \forall t \in \mathbb{R}, \quad \|R\|_{C_X,q}^q = \sum_{i=1}^m \phi(\sigma_i(R)) =: f(R)$$

Let $v(t) = \phi(t) + \beta(t-s)^2$ the proximal objective for some $s \geq 0$; we consider only non-negative s, as singular values are non-negative. As $v(t) \geq 0 = v(0)$ for $s = 0$, we note $Prox_\phi^\beta(0) = 0$. Observe that ϕ and therefore v are twice differentiable everywhere except at zero, hence by the chain rule:

- For $t \neq 0$: $\phi(t) = C_X|t|^q = C_X(\mathrm{sign}(t) \cdot t)^q \implies \phi'(t) = \mathrm{sign}(t)C_X q|t|^{q-1}$
  $\implies \phi''(t) = C_X q(q-1)t^{q-2}$
- For $t \neq 0$: $v(t) = \phi(t) + \beta(t-s)^2 \implies v'(t) = \mathrm{sign}(t)C_X q|t|^{q-1} + 2\beta(t-s)$
  $\implies \phi''(t) = C_X q(q-1)|t|^{q-2} + 2\beta$

For the proximal 1D-problems, we will apply the unconstrained CG-Newton method from the Python package scipy.optimize (cf. [11, p. 168]); gradients at zero are specified, although the function is not differentiable there, to return 0.

The Y-Subproblem

We rewrite the proximal mapping corresponding to the Y-subproblem by scaling by $\frac{1}{\beta} > 0$:

$$\arg\min_Y \ \lambda\|Y\|_1 + \beta\|Y - A\|^2 = \arg\min_Y \ \frac{\lambda}{2\beta}\|Y\|_1 + \frac{1}{2}\|Y - A\|^2 = \mathrm{prox}_{\frac{\lambda}{2\beta}\|\cdot\|_1}(A)$$

The proximal mapping of the l_1-norm for vectors is well-known and is calculated by elementwise soft-thresholding (see e.g. [4, Sect. 4.4.3]). As there is a bijection between tensors and vectors, and the l_1-norm of both representations are the same by its elementwise separability, the elementwise closed-form expression also holds for tensors. Hence, setting elementwise for any $t \in [T]_{\geq 1}$, $m \in [M]_{\geq 1}$ and $n \in [N]_{\geq 1}$

$$Y_{t,m,n}^* = \left(1 - \frac{\lambda}{2\beta \cdot |A_{t,m,n}|}\right)_+ \cdot A_{t,m,n} = \mathrm{sign}(A_{t,m,n}) \cdot \max\{|A_{t,m,n}| - \frac{\lambda}{2\beta}, 0\}$$

leads to an optimal solution, i.e. $Y^* = \arg\min_Y \ \lambda\|Y\|_1 + \beta\|Y - A\|^2$.

The Z-Subproblem

One observes that the Z-subproblem is a smooth, strongly convex program, therefore by setting the gradient to zero and solving for Z, we get the optimal solution Z^* of the proximal problem evaluated at A:

$$0 = \nabla_Z\left(\mu\|\cdot\|_F^2 + \frac{\beta}{2}\|(\cdot) - A\|_F^2\right)(Z^*) = 2\mu Z^* + \beta(Z^* - A)$$

Solving for Z^* yields:

$$Z^* = \frac{\beta}{2\mu + \beta}A$$

6.4 Final Algorithm

Collecting our considerations, we get the following algorithm that allows parallelization for the X-subproblem:

Algorithm 5 The Quasi-Stationary Proximal ADMM for Object Detection

Initialize $X^0, Y^0, Z^0, W^0 = -2\mu Z_0$
while stopping criteria not fulfilled **do**
  Determine X^{k+1}:
    Let $A_X^{k+1} = \frac{1}{2}(M - Y^k - Z^k - W^k/\beta + X^k)$
    Compute DFT of A_X^{k+1} by FFT along time-axis: $\bar{A}^{k+1} \longleftarrow \text{fft}(A_X^{k+1})$
    for $t = 1, \dots, T$ **do**
      Compute SVD of $\bar{A}^{(t)}$: $\bar{A}^{(t)} = U\Sigma V^T$
      for $k = 0, \dots, \min\{M, N\}$ **do**
        $s_k \approx |Prox_\phi(\sigma_k(A))|$
      end for
    end for
    $X^{k+1} \longleftarrow U \cdot \text{Diag}(s_1, \cdots, s_r, 0, \cdots, 0) \cdot V^T$
  Determine Y^{k+1}:
    Let $A_Y^{k+1} = \frac{1}{2}(M - X^{k+1} - Z^k - W^k/\beta + Y^k)$
    $Y^{k+1} \longleftarrow \text{sign}(A_Y^{k+1}) \cdot \max\{|A_Y^{k+1}| - \frac{\lambda}{2\beta}, 0\}$
  Determine Z^{k+1}:
    Let $A_Z^{k+1} = M - X^{k+1} - Y^{k+1} - W^k/\beta$
    $Z^{k+1} \longleftarrow \frac{\beta}{2\mu+\beta} A_Z^{k+1}$
  $W^{k+1} \longleftarrow W^k + \beta(X^{k+1} + Y^{k+1} + Z^{k+1} - M)$
  $k \longleftarrow k + 1$
end while
return X^k, Y^k, Z^k, W^k

6.5 Numerical Simulation

We now apply Algorithm 5 given by the quasi-stationary proximal ADMM to a sample video from the Dragon Lake Parking Dataset of the Model Predictive Control (MPC) Lab at UC Berkeley to detect cars (cf. [14]) [1]. The video is transformed into grayscale.
The biggest computational efforts stem from building the SVDs and DFTs in the X-Subproblem. Hence, we choose to concentrate on a small parking crossing patch of the video and resize the frames into a 180×160-format instead of on the whole high-resolution video over 7 minutes and 32 seconds with 3840×2160-size frames. Moreover, we choose to discretize the video by only taking every tenth frame, and in total 32 frames. So in total, there are $32 \cdot 180 \cdot 160 = 921.600 \approx 10^6$ variables. The transformed video clip shows two cars one after another driving into a crossing in a parking space from the top; the first car turns to the left while the second one drives forward from their perspectives respectively.

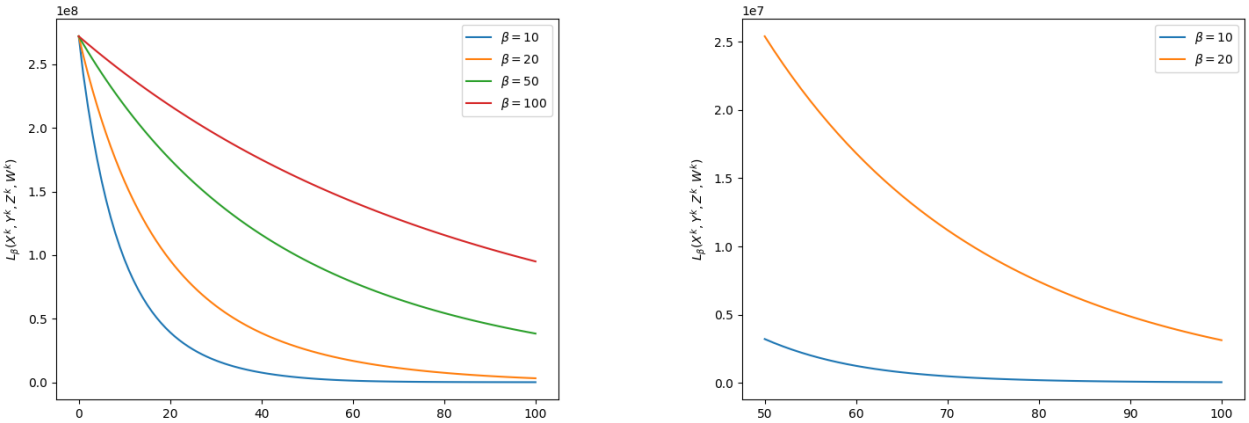
We choose the hyperparameters $q = 0.001$, $C_X = \sqrt{180 \cdot 160}$ (inspired by [20, Eq. (6)]), $\lambda = 20$ and $\mu = 0.5$ and simulate a run of 100 iterations for different values of $\beta \in \{10, 20, 50, 100\}$. The initial iterates are chosen as X^0 being the constant video where each frame is the average of all frames of the recorded video, Y^0 being the zeros-tensor, and $Z^0 := M - X^0 - Y^0$ such that the linear equality constraints are satisfied. The dual variable is chosen as $W^0 = -2\mu Z^0 = -\nabla h(Z^0)$ by the assumption in Algorithm 3, since $h(Z) = \mu\|Z\|_F^2$ in our model. As described before, we choose $H_X = H_Y = \beta \cdot \mathbb{I}_{T\times M\times N}$. For the X-subproblem, we use the solver-internal accuracy of scipy.optimize for the CG-Newton method applied to the 1D-problems.

[1] https://sites.google.com/berkeley.edu/dlp-dataset

The experiments are implemented in Python 3.12 and run on a laptop with Windows 11 that has an AMD Ryzen 5 8640HS 3.5GHz processor with six cores and 16 GB of RAM.

Influence of β on the Quasi-Stationary Proximal ADMM

Let us discuss the influence of β on the quasi-stationary proximal ADMM by regarding key quantities. We begin by plotting the augmented Lagrangian values in Fig. 6.1:



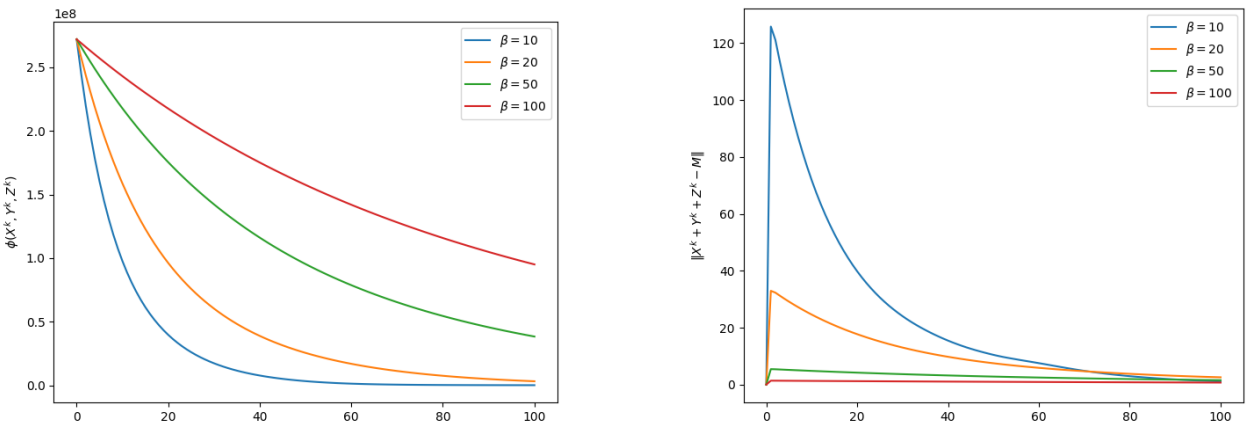
(a) Augmented Lagrangian values $\mathcal{L}_\beta(X^k, Y^k, Z^k, W^k)$ for iterations 0–100 and $\beta \in \{10, 20, 50, 100\}$

(b) Augmented Lagrangian values $\mathcal{L}_\beta(X^k, Y^k, Z^k, W^k)$ for iterations 50–100 and $\beta \in \{10, 20\}$

Figure 6.1 Augmented Lagrangian values for 100 iterations

The smaller β gets, the bigger the speed of descent is: For $\beta = 10$ the rate of decrease seems to be linear, while for the other tested values the rate seems sublinear (cf. Fig. 6.1a). The behavior for $\beta = 100$ is the slowest.
Since the scale of the augmented Lagrangian values gets much smaller for $\beta \in \{10, 20\}$ compared to the values for $\beta \in \{50, 100\}$, we consider the behavior again for particularly $\beta \in \{10, 20\}$ after 50 iterations in Fig. 6.1b. The continuous descent is evident, but since the scale in Fig. 6.1a is smaller than in Fig. 6.1b, we infer that the rate of change stagnates with increasing iterations.

Consider the objective values and feasibility violation in Fig. 6.2:



(a) Objective values $\phi(X^k, Y^k, Z^k)$

(b) Feasibility violation $\|X^k + Y^k + Z^k - M\|$

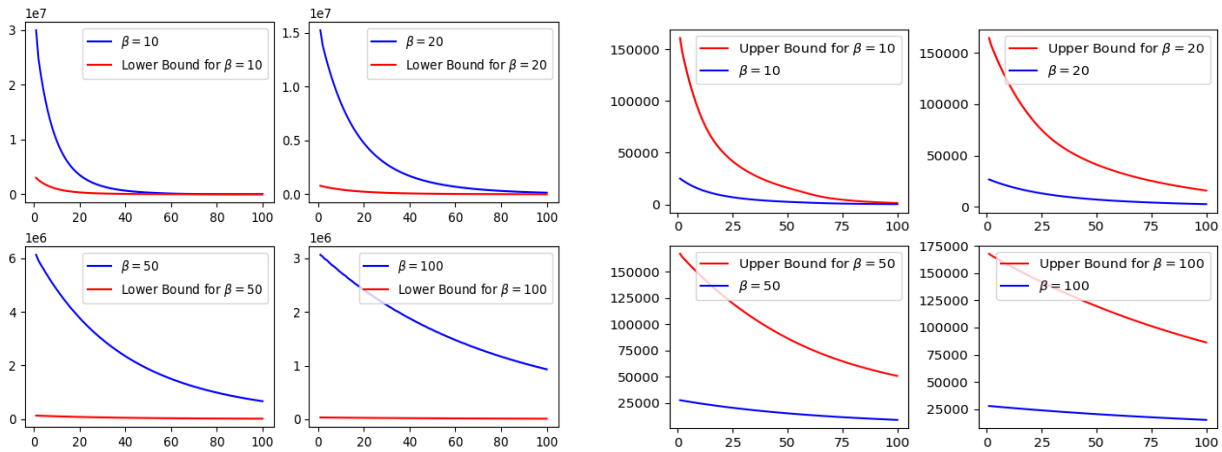
Figure 6.2 Objective values and feasibility violation

Comparing Fig. 6.2a with Fig. 6.1a, we see that the behavior of the augmented Lagrangian and of the objective values are similar, independent of the chosen value of β. This also means here: The smaller β is,

the bigger the objective descent over the whole run gets (cf. Fig. 6.2a). The similarity of Fig. 6.2a with Fig. 6.1a is due to the magnitude of the feasibility violation being much smaller, maximally reaching around 130 (see Fig. 6.2b), and the tested values of β being at most on the order of 10^2. On the other hand, the objective and augmented Lagrangian values are on the order of 10^8 in Fig. 6.1a and Fig. 6.2a, and therefore, they are approximately equal as the effects of the extra terms in the augmented Lagrangian are comparatively insignificant.
For each choice of β, after the first iteration, there is a jump in the feasibility violation visible in Fig. 6.2b. This can be explained by our choice for the initial iterates which satisfy the constraints exactly. For smaller β, the jump is higher, which can be attributed to the weaker penalization of the constraints. After a few iterations, there is a rising trend in fulfilling the constraints for all tested values of β as indicated by Fig. 6.2b.

So in our example, the choice of β dictates the objective-feasibility trade-off in the short term and especially afterwards tends to influence the long-term behavior of the objective function descent; however the constraints are approximately satisfied in all cases.

At last, let us plot the descent and subgradient norms along with the theoretical bounds P2 and P3 in Fig. 6.3:



(a) $\mathcal{L}_\beta$-Descent for 100 iterations **(b)** Subgradient norm for 100 iterations

Figure 6.3 The course of descent and subgradient norm (blue) compared to the theoretical bounds (red) for 100 iterations

We observe that the bounds are very loose initially and emerge closer with increasing iterations (cf. Fig. 6.3). Especially for smaller β, the deviation gets tighter, while for bigger β, there remains a discrepancy. The red lines, that represent the bounds from P2 and P3, are scalings of the residual norms. These theoretical bounds decrease faster for smaller β, even though in that case the residual norms are weighted weaker for our choice of the proximal terms in the subproblem costs. On the contrary, for bigger β, the residuals converge slowly.

Altogether, our observations in this specific example indicate that smaller β-values are better for prioritizing objective descent and residual convergence, whereas bigger β-values are better suited for faster constraint satisfaction. This indicates that an adaptive rule with an upper limit for the penalty parameter β, depending on the primary aims of the user, like in [20, Eq. (16)], might lead to better performance.

Performance in Object Detection

Let us briefly observe how well the algorithm performs for the use case of object detection. Since the application's primary aim is to detect the object's movement and not to obtain the exact background and noise, we use the small value $\beta = 10$ to focus on objective descent as discussed before. To evaluate the

performance, we mark the non-zero entries in the algorithm's foreground in red within the original video. Fig. 6.4 shows a screenshot of the marked videos, each corresponding to 50 and 500 iterations.

(a) After 50 iterations

(b) After 500 iterations

Figure 6.4 The detected vehicles by Algorithm 5

The algorithm detects the vehicles after 50 iterations (cf. Fig. 6.4a), despite our simplified model consisting of only constant weights and no adaptive rule for the penalty parameter β.

However, with increasing iterations (see Fig. 6.4b), the algorithm detects that our simplified optimization problem can be solved in a trivial manner by setting the total video as the background and neglecting the foreground and noise, which leads to a smaller objective value.
We explain this phenomenon in the following. As the matrix Schatten q-norm to the q-th power approximates the rank and $q = 0.001$ is small, each time slice of any input tensor from $\mathbb{R}^{32\times180\times160}$ will have Schatten q-norm of around $\min\{180, 160\} = 160$. In that way, by the definition of the tensorial Schatten q-norm as the average of the Schatten q-norm values of the time slices of size 180×160 in the Fourier domain, one can achieve an objective value around 160, since the l_1-norm and squared tensorial Forbenius norm are zero. This cost is comparatively low.

So although the quasi-stationary proximal ADMM regularizes proximity to the previous iterates with the proximal terms and theoretically, the residuals vanish in the limit, it has low bias towards previous iterations and detects the possibility of lower function values by choosing the trivial foreground.

7 Conclusion

Driven by the need for theoretical guarantees about ADMM in nonconvex nonsmooth optimization, the goal of this thesis has been to extend convergence results to general function classes while allowing inexactness by considering a modified variant of ADMM. For this purpose, we have extended an existing framework (cf. [17, Prop. 2]) to cases where certain conditions are not met, but the violation can be controlled by tolerances. The framework includes convergence results including asymptotic stationarity and sublinear rates of some convergence measures for summable errors. Additionally, we could infer global convergence of the iterates under suitable conditions.

Based on this framework and inspired by ideas from [1], [17], [19], we have developed and analyzed the quasi-stationary proximal ADMM, an inexact variant of the proximal ADMM. In order to do so, we have used tools from [13] for nonconvex nonsmooth optimization, like subgradient calculus and optimality conditions, to precisely formulate statements and obtain a profound analysis. We have shown that under certain assumptions, the regular ADMM is a specific instance of the quasi-stationary proximal ADMM and continued to recover the convergence results from the framework. At last, we have considered a nonconvex nonsmooth model for object detection. We have shown a typical workflow for configurating the algorithm and solving the subproblems, giving efficient schemes in comparison to black-box global solvers which would be computationally infeasible. The process has shown us that ADMM needs to be tailored to the specific problems, underlining the importance of good problem formulations as well as individual considerations specific to the model.

The method applies to separable nonconvex continuous optimization problems with linear constraints, however other constraints could only be included by adding the indicator function to the cost. Hence, one open research direction might be to examine ways to enlarge the cost and constraint function classes.
Additionally, our numerical experiments indicate that an adaptive penalty parameter β might lead to better performance in comparison to fixed choices, to retain steep objective descent while the constraints are satisfied quicker. At the same time, we have observed stagnation of the augmented Lagrangian descent and subgradient norms after some time. Therefore, one big research direction one can investigate are further strategies to accelerate the algorithm, including combinations with faster solvers.

In total, the developed algorithm can be applied to very general problems with theoretical guarantees, and hence has much potential in today's world with the increasing use of nonconvex models. Simultaneously, it stimulates new research questions, especially examining strategies to accelerate and extend the algorithm to more general cost and constraint classes.

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